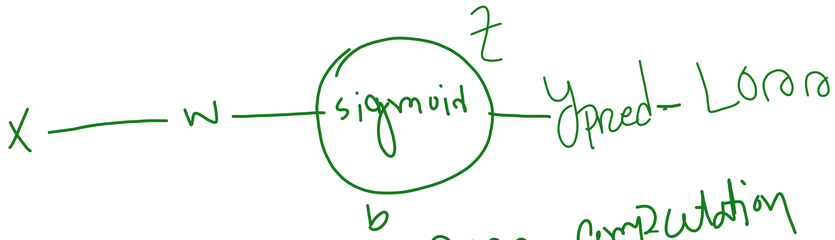
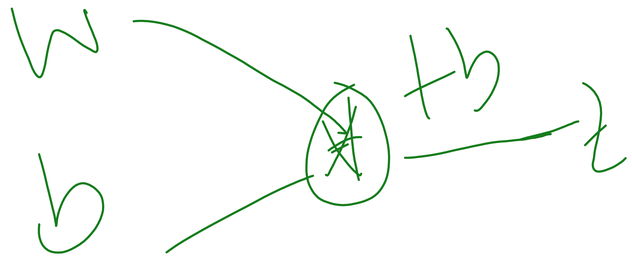
Autograd for neural network

Study hours	Sleep	pass/fail
2	3	0
5	6	1
7	8	0
9	10	1

[Training process]





1) forward pass
2) Calculate loss
3) Backward pass
4) update gradients

forward pass computation

1. Linear transformation

$$z = W \cdot X + b$$

2. Activation (sigmoid f^n)

$$y_{pred} = \sigma(z) = \frac{1}{1+e^{-z}} = \hat{y}(1-\hat{y})$$

3. Loss f^n

prediction = 0, 1

$$L = -\left[y_{target} \cdot \ln(y_{pred}) + (1-y_{target}) \cdot \ln(1-y_{pred})\right]$$

derivative

der.

$$\frac{\partial L}{\partial w} \quad , \quad \frac{\partial L}{\partial b}$$

$$\frac{\partial L}{\partial y_{pred}} \times \frac{\partial y_{pred}}{\partial z} \times \frac{\partial z}{\partial w}$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial y_{pred}} \times \frac{\partial y_{pred}}{\partial z} \times \frac{\partial z}{\partial b}$$

$\frac{\partial L}{\partial y_{pred}}$

$$L = -\left[y \cdot \ln(\hat{y}) + (1-y) \cdot \ln(1-\hat{y}) \right]$$

$$= -y \ln(\hat{y}) - (1-y) \ln(1-\hat{y})$$

$\boxed{\ln(x) \cdot x}$

1st part

$$\frac{\partial}{\partial \hat{y}}\left(-y \ln(\hat{y})\right) = -y \cdot \frac{1}{\hat{y}} = \frac{-y}{\hat{y}}$$

2nd part:

$$-(1-y)\ln(1-\hat{y})$$

$$+(1-y) \cdot \frac{1}{1-\hat{y}} = \frac{(1-y)}{(1-\hat{y})}$$

$$= \hat{y} - y$$

$$\frac{\hat{y}}{\hat{y}(1-\hat{y})}$$

Answer

$$\frac{\partial z}{\partial \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

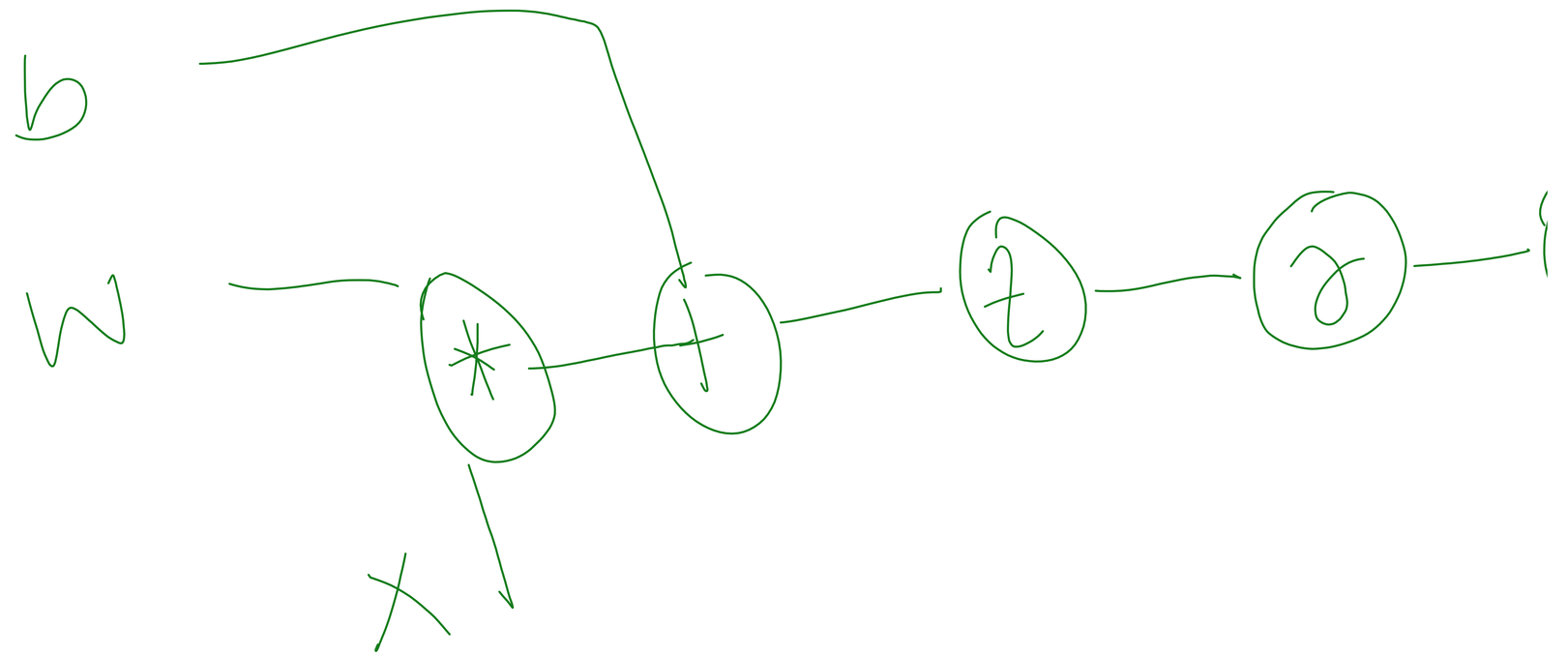
$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1-\hat{y})$$

$$\frac{\partial z}{\partial w} = x$$

$$T^*(R - h) = \frac{ae}{7e}$$

$$T^*(R - h) = \frac{ae}{7e} \qquad \#$$

$$T = \frac{ae}{7e}$$

Computation graph

requires graph